

Exploring the Collaboration Possibilities of Distributed Making for Storytelling Using 3D Printing Pens

Amina H. El-Ashry, Xinran Zhang, Savani Shrotri, Susanna Abler, Foad Hamidi

Information Systems Department, University of Maryland, Baltimore County (UMBC),

{aminaelashry,x16,savanis1,sabler1,foadhamidi}@umbc.edu

ABSTRACT

As digital making tools, such as 3D printers and laser cutters, become more affordable and easier to use by non-experts, new opportunities arise for geographically distributed designers to use them collaboratively and at a distance. In this exploratory study, we investigated the possibilities of a 3D printing pen for collaboratively creating multimodal stories around the themes of ecology and climate change for elementary school-aged children. Our interdisciplinary team of four students met regularly over a video link and used shared documents and 3D pens to create a multimodal adventure book, with audio and tactile elements, that we then used as a design probe to elicit feedback from experienced community science educators working with children about the possibilities of using hands-on distributed collaborative techniques in the future. We found that using this approach is promising for facilitating creative verbal and visual collaboration at a distance, and using tactile and audio representations with narrative elements provides new ways to reflect on complex topics, such as climate change.

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While the idea of remote collaboration using digital tools, such as video links or shared documents, is not new (e.g., [8, 27]), in the last few years, there has been a proliferation of online platforms (e.g., Zoom, WebEx, etc.) that connect geographically distributed teams to each other and offer opportunities for collaborative brainstorming, designing, and making activities [10]. In addition, the need to move educational and creative tasks online in response to the COVID-19 pandemic in 2020 and 2021 also accelerated this move to online and hybrid modes of collaboration (e.g., [19, 24]). While historically, digital fabrication and making activities, including those for children, have often taken place in a shared physical space (e.g., a makerspace, fabrication lab, or afterschool learning site, among others) [3, 21], there is a need to better understand the possibilities of distributed approaches to collaborative making. This

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project contributes to this research space by exploring how we can leverage remote collaboration techniques for creating interactive design artifacts, such as tactile storybooks (Figure 1).

In this exploratory project, we studied the following research question: what are the possibilities and limitations of using a combination of video, shared documents, and craft and 3D printing tools to create interactive multimodal artifacts for children collaboratively? We used 3D pens in this project because compared to conventional 3D printers (that we had difficulty accessing in our research lab due to the pandemic), they are easier to distribute in users' households, according to manufacturers, are safer to us by non-experts with a decreased probability of catching fire or causing burns, and are designed for use in combination with craft materials [28]. 3D pens (Figure 1, top right) are handheld devices that work similarly to consumer-grade additive 3D printers by melting plastic inside that is then extruded from the pen's tip. The pen we used has a slot at the top end to insert plastic filament (i.e., PLA), which it then heats, melts, and extrudes from the other end. The user can then move the tip in a desired direction to create forms and shapes from layers of plastic.

The storyline of our interactive storybook consists of an animal character traveling around the world through four habitats (each made by a different team member). The reader interacts with tactile and audio elements in each habitat (Figure 1, left and middle), such as characters (Figure 1, bottom right). We decided to create interactive multimedia interfaces for children using our approach because this study is part of a longer-term research project that studies the potential of making and digital fabrication for community engagement in the context of informal learning, with a focus on youth and children. We found the combination of video sessions, shared documents, and making tools effective for facilitating collaboration. Next, we present a summary of related work before describing our approach and initial feedback from a community partner on the project. Finally, we conclude with a discussion of future directions.

1 RELATED WORK

Remote collaboration has been a common method for sharing ideas and working on projects for designers and makers. It has proved to be an effective method with many benefits to performance and streamlined work [20]. Techniques of observing remote behaviors on-screen help one understand the mannerisms and read the person in front of them, which is essential when collaborating remotely, providing more natural human communication. Some of these techniques are camera positioning [22], visible as well as non-visible gestures [9], and even the type of device being used for remote communication [17].



Figure 1: A jungle habitat with trees and branches as obstacles that need to be traversed by a character (left); another jungle habitat with a river and water as obstacles for the snake character to move through (middle). Both pages have an audio module that plays ambient jungle sounds. The 3D pen used in the project (top, right); and a jungle habitat character, the Snake (bottom, right).

A significant amount of research on creating tactile media using maker approaches has shown that multimodal interfaces, including those combining tactile and audio components, can support collaboration and exploration for a wide range of individuals, including children [5, 12, 14, 15, 26]. Past research has explored several possibilities for these tactile interfaces, including creating tactile games [1, 12, 14] or puzzles [12, 23] that were paired with an electronic device. For example, Hu et al. developed a tangible iPad app called *Strawbies* to introduce children to computational thinking [14]. The user interacts with the app by harvesting and protecting strawberries, constructing wooden puzzle-like tiles to simulate the movement on the screen using image recognition technology. During evaluations, researchers saw that younger and older children could use the images on the tile to associate with movement on the screen and create more complex paths using the provided tiles, allowing for this desired flexibility. In another project, Kang et al. developed a system, *PrototypeAR*, that combined papercraft, AR, and online simulation to support the collaborative creation and testing of low-fi models of complex systems, such as bicycles or cameras, by child users with a combination of low-fi prototyping materials, AR technology, and software simulation [18]. They observed several collaborative behaviors in the users, including splitting design tasks and discussing ideas and feedback. Similarly, Hamidi et al. used a low-fi prototyping approach to create interactive models of the environment with children collaboratively and found that the shared tangible nature of activities was engaging for children in Mexico, and it allowed for incorporating local materials and ecological information into designs [11]. Previous research has shown that interactive multimedia interfaces can engage diverse users and encourage collaboration and creativity. Our study adds to this body of work by exploring distributed ways that these interfaces may be produced.

2 EXPLORING DISTRIBUTED MAKING A MULTIMEDIA INTERACTIVE TACTILE BOOK

To explore the possibilities and challenges of using a distributed making approach to create an interactive tactile storybook collaboratively, our interdisciplinary research team participated in 12 remote brainstorming sessions (over the WebEx video platform), each lasting approximately 60 minutes and complemented with individual prototyping activities for three and a half months. Our team consisted of two undergraduate students (one majoring in Information Systems and the other in Fine Arts), two graduate students (both in Information Systems), and a faculty research advisor. Our goal from these sessions was to identify and use practical approaches to designing a tactile narrative using 3D pens, specifically a commercial product called 3Doodler [28]. Each research team member had a 3Doodler pen and a set of multicolored PLA plastic spools that they could use to create prototypes. Over the course of the project, we used a shared Google Drive folder to share documentation and meeting notes.

2.1 Brainstorming

At the outset, we had to figure out two key aspects of the project: the role of the 3Doodler pen and the structure of the tactile narrative that was going to be engaging its reader through multimedia elements. We first familiarized ourselves with the 3D pen using instructional videos and documentation provided by the manufacturer. Next, we each explored the use of the pen on our own and shared our experiences about the pen's capabilities and challenges through discussions over video and affinity mapping. Creating small "test" objects before designing the tactile narrative pages proved helpful for quickly trying out alternatives and communicating with other team members by showing them ideas. For example, this exploration helped us gain familiarity with the texture of the 3Doodler filament, its flexibility, the amount of time it takes for the filament

to harden, and how fast the filament is extruded, to which surfaces it sticks on, among others. The instructional videos also provided basic information on how to use the pens safely and avoid making loose or sharp extruding pieces to make sure designed artifacts are safe for children to use.

With this information, we moved onto structuring the tactile narrative. Questions during this phase included: What storyline would be educational and engaging? What types of fabrication tools would result in tangible elements that children may recognize and enjoy using? How can we incorporate audio components into the narrative to make it more engaging? Inspired by previous research on informing children about complex societal and environmental issues (e.g., [4, 11]), we focused on a story about climate change. The story would explain climate change in a simple manner that shows children how different habitats could be affected and provides us with various environments with other tactile elements to experiment with. As described previously, each of our team members created one of four habitats (i.e., arctic, tundra, jungle, and desert) as a mini-challenge that the reader has to traverse to get to the next stage as part of the whole experience.

2.2 Prototyping the Narrative

Each team member was tasked with prototyping low-fidelity and high-fidelity versions of the storyline's habitats, from the desert to the jungle. Three iterations of each page in total were implemented, and at the early prototyping stages, we received internal feedback from our research advisor and each other. Later on, we received external feedback, which will be discussed shortly. In low and high-fidelity prototyping, four elements were essential to designing engaging and clear pages: tactile element placing, spacing between elements, interaction types with the page, and crafting materials used.

We anticipated that the location of the tactile element affects the attention a reader may give it and the order in which it is explored. Therefore, we found it important to make sure elements that depend on other elements are in the middle of the page to attract the most attention and start a domino effect of exploration in the correct order. For example, in the jungle habitat page of the tactile narrative (Figure 1, left), 3D-designed trees are the largest and placed in the middle of the page. The pins on the trees are the core elements from which interaction is done and encourages users to explore other parts such as the clouds, birds, grass, and flowers, as shown in Figure 2

We decided to create space between tangible elements to make it potentially easier for more than one person (e.g., two children or a child and a caregiver) to explore the story elements. We used different interaction styles for each habitat to make the activities engaging and diverse, from magnetic attachments and action-triggered audio to grooves in the cardboard pages to simulate sliding movements. We used MultiPlay [16] audio modules to play ambient sounds during interaction to increase engagement.

Previous research on tactile interfaces has emphasized including elements that help users feel part of a story and play a meaningful role in its unfolding progression [1, 5, 7]. We addressed this by including a range of interactive elements (e.g., allowing users to navigate through a short maze or move characters from page to



Figure 2: Side view of the jungle page and its tactile elements.

page) in the book. We used different types of craft materials to correspond to the element being represented. For instance, we used felt to represent greenery, cotton balls to represent clouds, pom-poms to represent flowers, etc. We used the 3D pen's plastic material to represent concrete and important interactive elements, such as trees and characters.

A limitation of our approach was that we could not touch the tactile elements of each other's prototypes over video. Team members had to listen to descriptions of the elements rather than experiencing them firsthand. A possible solution to overcoming this limitation is to use a hybrid format where team members can occasionally meet and interact with prototypes in the presence of safety measures. Another important limitation of our approach was that during the initial phase of the project, much of our brainstorming took place within the project team. This was mainly due to the COVID-19 pandemic while limited opportunities for engaging with stakeholders directly. In the subsequent steps of our project, we mitigated this limitation by working remotely with a community partner (described next).

3 PRELIMINARY COMMUNITY FEEDBACK

Once our high-fidelity prototyping was complete, we shared the work with an afterschool climate change and bird ecology awareness community group in the local area. The engagements involved both children and adult educators. Our goal for these engagements was to receive early formative feedback on the prototypes to figure out if the design concept was clear and appealing to others.

In a first informal session of about 60 minutes, we showed the prototype via video link to 8-10 late elementary and middle school children who participated in the program and three adult educators. This session aimed to establish rapport with the group and understand general reactions to the prototype from different stakeholders. Our protocol consisted of a round of introductions, a demonstration of the tactile narrative pages and interactions, and time for the group to provide feedback and ask questions. During this session, we did not collect data from specific individuals, as we did not have the opportunity to gather all parental consent needed. Nevertheless, the experience allowed us to reflect on the design that resulted in

some modifications, including adding more actions for a user to perform on pages of the narrative and experimenting with different craft materials to make representations clearer.

In a second session, held approximately three months after the initial session and conducted via video link, we conducted a focus group with the three adult educators who had participated in the first session. The purpose of this session was to understand their perspectives on our remote making approach and techniques rather than specific design decisions related to the interactive book. The educators consisted of an experienced avian biologist (P1), a Human-Computer Interaction PhD student (P2), and a Sociology PhD student (P3). The participants conducted climate change awareness sessions in a virtual format due to the COVID-19 pandemic. We used a focus group protocol to ask participants about their perspectives on the remote making approach and suggestions for future steps. The session took approximately 60 minutes, and participants received \$25 for their time. Our university's IRB office approved the study. We recorded and transcribed the session and conducted a deductive thematic coding where we identified topics that corresponded to our research themes of collaboration and remote making. We asked the participants to reflect on and share ideas for using interactive design activities to engage youth and children throughout the session.

Our participants acknowledged the importance of collaboration for building youth's confidence when participating in learning and making activities. They described how these activities should be designed to make the users feel like they have accomplished something by adding to their skill set or confidence in their abilities. For example, P3 said, "As a team, as a group, [they reflect] on something that they can share between each other. Like, 'Yeah, I did that.' And then they're talking to each other and relating to their experience in that." Previous research has shown that making and fabrication activities can lead to creative expression and self-efficacy in children (e.g., [6, 9]), and combining these types of activities with climate change awareness can create a balance where individuals are both informed about a complex topic but also possibly find the creative process rewarding.

Furthermore, reflecting on their community-engaged activities that often involve Spanish-speaking participants, the educators expressed that hands-on physical activities such as crafting or games, when done in-person or virtually in a collaborative manner, can be meaningful and inclusive for people who may experience language barriers. For example, P2 said, "[Hands-on prototyping] seems to engage [everyone], maybe even with the language barrier, [they] can still kind of see what the others are doing physically [and play along]." Compared with others that focus only on verbal or written communication, these activities can allow more people to express themselves and contribute to the group's activity. Participants also described how collaborative activities with playful and fun elements engaged the youth they work with effectively. For example, P3 said, "[when] there is any sort of fun that is to be had on a group level, they usually take away a lot more in those instances than if [one was to] do individual activities." At the same time, participants also stated the importance of breaking down collaborative activities into practical and straightforward tasks that individuals can complete to contribute to the group's overall success. For example, P1 gave an example of this with the middle schoolers participating in the

climate change awareness program: "We did one activity last semester where [middle schoolers] go birding, and they take notes about what they see. And they go in small groups. So, it's kind of a group activity, but it's also like an individual activity because they each hopefully have their own binoculars and their own notepad. And then it becomes a group activity later when they have to choose among themselves who will report a story about something they saw. This led them to find their voice just within the people that they go to school with. And then giving them this knowledge and them being able to say it out loud and be a leader about it." This example is relevant to our collaborative experiences as a design team when working on the tactile narrative: While we collaborated on brainstorming and prototyping, our individual making experiences were essential to the streamlined completion of the overall design artifact – the tactile narrative.

4 CONCLUSION AND FUTURE WORK

In this study, we explored the possibility of using a remote distributed approach to fabricating multimodal tactile books using 3D pens. We found that combining video sessions, shared documents, and easy-to-distribute making tools provide effective ways to facilitate interdisciplinary collaboration. Additionally, our community partners identified the potential of using distributed making to creatively explore concepts related to ecology and the environment in the future.

A key limitation of our project is that we did not collect feedback and data directly from children. In the future, we plan to engage children in working with the fabrication tools that we explored and involve more community members in our research activities (usability sessions, focus groups, interviews). Furthermore, given that our communication was mainly video-based, a limitation was that we could not touch and feel the tactile elements each of our team members was designing. This required us to document our progress via photos and videos, with detailed verbal explanations of how surfaces felt to the touch. In the future, using hybrid approaches that combine some in-person interaction with remote activities would be interesting to explore. Furthermore, using Virtual or Augmented Reality tools in combination with the making techniques can effectively facilitate creativity. Finally, given that tactile books may be more accessible to users with visual impairments, in the future, we would like to study the potential of our approach for enabling novel co-reading activities [25] for parents and children with diverse visual abilities.

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